

DEMO • PRODUCT REVIEW



Saudi MuayThai Federation

Membership, Live Scoring & Digital Accreditation Platform

STACK

.NET 8 · React 18 · Flutter · SQL Server · SignalR

ARCHITECTURE

Clean / CQRS · Outbox · HMAC Tokens

BUILD

One-command Docker Compose

The Problem

Federations run on spreadsheets, paper IDs and chat groups. It does not scale.



Fragmented operations

Member files in Excel, fees collected by bank transfer, referee assignments by WhatsApp, scores written on paper at the venue.



No source of truth

Who is registered? Paid? Medically cleared? Eligible to compete tonight? Every question requires a phone call.



Trust gaps at the gate

Printed cards are forged or shared. There is no offline-capable, verifiable identity at the venue door.



Zero live experience

Audiences cannot follow a bout in real time. Sponsors cannot point at engagement numbers. Brackets are PDFs.

Our Solution

One platform. Three surfaces. Every federation workflow under a single source of truth.

W

Public Web

Member registration, fee payment, club microsites, news, live tournament brackets and live scoreboard for spectators.

React 18

Vite

SignalR

A

Admin Panel

Approvals, scheduling, referee assignment, head-referee & timekeeper consoles, score overrides, certificates, exports.

Role-based

Audit log

M

Mobile (Flutter)

Athlete digital ID with offline QR, EN/AR + RTL, auto-refresh on connectivity, gate-scanner-ready.

Offline-first

HMAC-signed

Backend

.NET 8 minimal-API, MediatR CQRS, FluentValidation, EF Core, SignalR hubs.

Reliability

Transactional outbox, idempotent webhooks, optimistic concurrency on ID issuance.

Deploy

One `docker compose up` brings up DB + API + Web + Admin.

Architecture

Clean Architecture + CQRS — a stable core surrounded by replaceable I/O.

Clients

Web SPA · Admin SPA · Flutter

Edge

Nginx reverse proxy
HTTPS / WebSocket

Application

.NET 8 API · MediatR
SignalR Hubs

Data & Async

SQL Server · EF Core
Outbox poller

Why Clean Architecture

- **Domain** has zero external dependencies — pure entities & invariants.
- **Application** orchestrates: commands, validators, handlers.
- **Infrastructure** is swappable: EF Core today, anything tomorrow.
- **API** is a thin host that maps HTTP/JSON to commands.

Why CQRS + MediatR

- Each business action is a single explicit **Command**.
- Pipeline behaviors give us validation, logging, retries for free.
- Read models can evolve independently from writes.
- Testable end-to-end without spinning up the web host.

MODULE 01

Membership Registration

From sign-up to a verified, paying, medically-cleared SMF member.

Membership Lifecycle

A single command-style endpoint, defended in three layers.

Smart, generated SMF ID

Format `#SMF{YYYY}-{00001}`. Allocated atomically with optimistic concurrency, so two sign-ups at the same instant never collide and never skip a number.

Defense in depth

- UI hides irrelevant fields (Zod live validation).
- Server re-validates with FluentValidation.
- Domain entity enforces invariants on `Member.Register()`.

Minor athlete protection

If *Role* = *Athlete* and age < 18, **Guardian Consent** is required — checked in UI, schema, server validator, and entity.

Medical clearance & approvals

Admin records clearance, approves or rejects. The state machine is on the entity itself, not scattered across handlers.

MODULE 02

Payments & Outbox

Take money safely. Never lose a receipt. Never double-issue an ID.

Payments — Reliability First

Webhooks are unreliable by nature. We design for exactly-once business effects.

Signed webhooks

Every payment provider callback is verified with HMAC against a shared secret. Forgeries are rejected at the door.

Idempotent state transitions

`Payment.MarkSucceeded()` is a no-op on duplicates. Replaying a webhook ten times produces the same single fee record.

Transactional outbox

`PaymentSuccessfulEvent` is staged in the same SQL transaction that flips the payment row. A background poller dispatches it through MediatR with bounded retries.

Atomicity across processes

Crash mid-handler? The outbox row stays pending and the next poll picks it up. **No event is ever lost.**

MODULE 03

Offline Digital Accreditation

One QR code. Works without internet. Cannot be forged. Revocation-aware.

Digital ID at the Gate

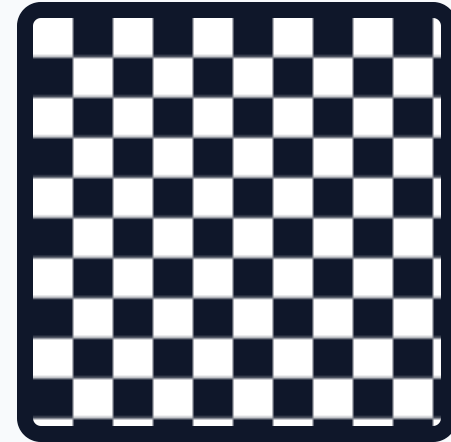
How it works

1. **Issue** — server signs an HMAC-SHA256 token:
`base64url(payload).base64url(sig)`.
2. **Cache** — Flutter stores it in Hive; QR renders on first paint, no spinner.
3. **Scan** — gate device scans, calls `/admissions/verify`.
4. **Verify** — server re-checks signature *and* live `RegistrationStatus`.
5. **Refresh** — app silently re-issues at the offline→online edge.

HMAC-SHA256

iat / exp

Bilingual EN/AR + RTL



#SMF2026-00042 — Approved

Token valid · 24h · revocation-aware

Gate Decisions — Every Outcome Is Explicit

All denials are HTTP 200 on purpose, so dashboards distinguish "scanner offline" from "token rejected".

Outcome	HTTP	What it means at the door
Admitted	200	Signature valid, in window, status Approved or Active.
TokenMalformed	200	Cannot parse <code>payload.sig</code> . Likely a non-SMF QR.
SignatureMismatch	200	Forgery, replay across environments, or rotated signing key.
NotYetValid / Expired	200	Outside <code>iat</code> / <code>exp</code> window (clock skew honored).
MemberUnknown	200	Token references a member id that no longer exists.
MemberNotInGoodStanding	200	Live status is Pending / Suspended — denied even with a valid token.

MODULE 04

Live Scoring & Tournaments

Real-time bouts, live brackets, audited overrides — for fighters, referees and audiences.

Real-Time Scoring Pipeline

SignalR hubs broadcast every point, override and timer tick to every connected client.

R

Referee console

Side-judge taps a score; it appears on the public scoreboard in < 100 ms.

H

Head referee

Can override a panel score. Every override is persisted as an audit event with reason & actor.

T

Timekeeper

Round, break and total clocks broadcast over the same hub. Reconnect-resilient.

Tournament brackets

Auto-generate a single-elimination bracket from approved registrations. Public bracket page renders connectors live as winners advance.

Replay & analytics

Every score event is event-sourced — fighters get a per-round breakdown, federation gets engagement and accuracy metrics.

MODULE 05

Public Engagement & Operations

Clubs, news, certificates, live streams, exports — the whole federation in one place.

What's Already Built

A non-exhaustive look at the modules shipping in the demo today.

Members

Registration · approvals · medical clearance · digital ID issuance · status timeline.

Payments

Gateway integration · signed webhooks · outbox · receipts.

Matches

Schedule · referee assignment · live scoring · score overrides · replay.

Tournaments

Bracket generation · live propagation · public bracket view.

Clubs

Club microsites · roster · contact & news.

Certificates

HTML-rendered, downloadable, signed certificates of participation.

News & Events

Federation news feed and event calendar with categories.

Live streams

Provider-agnostic embed (YouTube, Twitch, custom).

Admin & Analytics

Role-aware admin · dashboards · CSV/JSON exports · safeguarding case log · notifications.

SMF Demo

Technology & Quality

Backend

- .NET 8 · ASP.NET Minimal API
- MediatR · FluentValidation · EF Core
- SignalR for real-time hubs
- SQL Server 2022 · LocalDB / InMemory fallback
- Serilog structured logging

Frontend & Mobile

- React 18 + TypeScript + Vite + Tailwind
- react-hook-form + Zod end-to-end
- Flutter (Hive cache, EN/AR with RTL)
- Nginx static hosting + reverse proxy

DevOps

- Single `docker compose up` bootstrap
- Cloudflare Tunnel for instant public demos
- EF Core migrations, applied on Dev boot
- Secrets via .NET user-secrets / env / KMS

Testing & Hardening

- Unit + handler tests (xUnit + WebApplicationFactory)
- ProblemDetails error responses everywhere
- Idempotency · retries · concurrency tests
- CORS allow-list configurable per environment

Live Demo Flow

Five minutes, end-to-end. Works on the demo URLs at the back of this deck.

1 • Register

On the public site, register an athlete. Watch `#SMF2026-XXXXX` appear instantly. Try a minor without guardian consent — see the inline server error.

2 • Approve

In the admin, approve the new member. Status flips Pending → Approved. Federation now has a real, addressable member.

3 • Scoreboard live

Open the public scoreboard for a match. In admin, drive Scoring / Head Referee / Timekeeper consoles — the public page updates in real time.

4 • Bracket generate

In Brackets, click *Generate from registrations*. Open the public tournament page — the bracket renders with connectors as bouts complete.

5 • Digital ID

Issue a digital ID for the athlete. Scan the QR at the gate (or call `11 / Demo / admissions/verify`) — admitted, expired, or denied with a clear outcome.

6 • Resilience

Replay a payment webhook ten times — exactly one fee row. Kill the API mid-flow — outbox replays the event on restart. Nothing lost.

What's Next

Hardening the demo into a federation-grade production system.

Auth & SSO

OIDC + roles for athletes, referees, club admins, federation staff. Gate every issuance endpoint behind real identity.

Productionised infra

Managed SQL (Azure SQL / RDS) · TLS via Caddy/Traefik · log aggregation (Seq / Loki) · CI image builds.

Mobile maturity

Apple/Google account binding · screenshot prevention · push for fight-card & gate updates.

Horizontal scale

Outbox lease column for multi-instance dispatchers · Redis backplane for SignalR.

Federation analytics

Cohort dashboards · referee accuracy · revenue · sponsorship engagement metrics.

Internationalisation

Full Arabic translation pass across web/admin (mobile already EN/AR).

THANK YOU

Let's run the demo.

Public web · Admin · Swagger — live, on the URLs below.

Public web <https://cosmetic-step-slip-wrapping.trycloudflare.com>

Admin panel <https://calculated-entrance-helping-memphis.trycloudflare.com>

Swagger <https://cosmetic-step-slip-wrapping.trycloudflare.com/swagger/>

URLs are temporary cloudflared quick-tunnels and rotate on each run.